Gravitational Lensing and Early-Universe Behavior in the Causal Diamond Framework

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Abstract

We derive two predictions of the causal diamond framework for MOND that extend beyond galaxy rotation curves. First, gravitational lensing tracks dynamical gravity: the entropy deficit that produces the MOND-enhanced gravitational force acts as an effective stress-energy source, generating real spacetime curvature that bends light and deflects matter equally. This follows from the distinction between modified source (which the framework does) and modified action (which it does not). The framework thus predicts lensing = dynamics, consistent with galaxy-galaxy lensing observations. Second, MOND effects vanish in the early universe: in the matter-dominated era, the proper event horizon grows without bound, the volume-law entropy budget becomes unlimited, and the entropy balance reduces to pure Newton. This is the correct Einstein–de Sitter limit and is consistent with the success of standard gravity in CMB physics and structure formation. Together, these results show that the causal diamond framework is compatible with the two main observational contexts where standard MOND faces difficulties.

1. Introduction

The causal diamond framework [1, 2, 3] derives the MOND acceleration constant and interpolation function from the entropy structure of the observer's causal diamond in ΛCDM. Companion papers have established the gravitational wave speed prediction ($c_{gw} = c$ [4]), the redshift evolution of a_0 [5], and the connection to the cosmological constant problem [6]. Two important questions remain for the framework's viability: does the MOND-enhanced gravity produce gravitational lensing consistent with observations, and does the framework reduce to standard gravity in the early universe where the cosmic microwave background (CMB) and structure formation require it?

This paper addresses both questions. The answers follow from the internal structure of the framework without additional assumptions.

Part I: Gravitational Lensing

2. The Problem

In general relativity, gravitational lensing and gravitational dynamics arise from the same metric. A photon and a massive particle orbiting a gravitating body both respond to the same spacetime curvature. The "missing mass" inferred from rotation curves is also seen in gravitational lensing: galaxy-galaxy lensing measurements [7, 8] confirm that the lensing signal matches the dynamical mass, not the baryonic mass alone.

Modified gravity theories face a choice. If the theory modifies the metric differently for lensing and dynamics (as the original TeVeS did [9]), it can potentially decouple them. If it modifies both equally, lensing tracks dynamics. Observations require the latter, at least at galaxy scales.

The causal diamond framework, as presented in Ref. [4], does not modify the spacetime metric. This raises a concern: if the metric is unmodified, photons follow standard GR geodesics, and lensing would see only the baryonic mass—contradicting observations. We show below that this concern is resolved by the distinction between modifying the gravitational action and modifying the effective gravitational source.

3. Modified Source, Unmodified Action

The causal diamond framework modifies gravity through the entropy balance: the effective gravitational coupling is $\mu(x) = kx/(1+kx)$, reducing the fraction of degrees of freedom in the Newtonian channel at low accelerations [2]. The MOND equation $\mu(g/a_0) \times g = g_N$ then gives $g = g_N/\mu > g_N$ in the MOND regime.

This enhanced gravitational field must be sourced by something. In the entropy framework, the source is the entropy deficit: the volume-law dark energy entropy that has been “consumed” by the transition from Newtonian to MOND dynamics acts as an effective stress-energy contribution. Following Verlinde [10], this entropy displacement produces real spacetime curvature—it is an effective gravitational source, not a modification of the propagation equations.

The distinction is critical:

What is modified	Physical effect	Consequence
Action (kinetic term)	GW propagation speed changes	c_gw ≠ c (ruled out)
Source (stress-energy)	Spacetime curvature changes	Lensing = dynamics

Table 1. The source/action distinction. The causal diamond framework modifies the effective source (bottom row), not the action (top row).

Because the entropy deficit acts as a genuine gravitational source, it curves spacetime in the standard GR sense. Photons follow geodesics of this curved spacetime. The lensing signal therefore includes the entropy-deficit contribution, and lensing tracks dynamics:

$$\text{Lensing mass} = \text{Dynamical mass} = \text{Baryonic mass} + \text{Entropy-deficit apparent mass} \quad (1)$$

This is consistent with galaxy-galaxy lensing observations [7, 8] and with the radial acceleration relation measured via weak lensing [8].

4. Consistency with the Gravitational Wave Speed

The source/action distinction resolves what would otherwise be a contradiction. In Ref. [4], we showed that the framework predicts $c_{gw} = c$ because it does not modify the gravitational action. Here we show that lensing tracks dynamics because it does modify the effective source. Both statements are true simultaneously:

- The **action** (Einstein-Hilbert) is unmodified → gravitational waves propagate at *c*
- The **source** (effective stress-energy) is modified → spacetime curvature tracks the MOND-enhanced gravity
- Photons follow geodesics of the modified curvature → lensing = dynamics

This is the same physical picture as Verlinde’s emergent gravity [10], where the entropy displacement creates “apparent dark matter” that is gravitationally real: it curves spacetime, bends light, deflects matter, and produces gravitational waves that travel at *c*.

5. Implications for the Bullet Cluster

The Bullet Cluster (1E 0657-56) is widely regarded as the strongest evidence for particle dark matter [11]. Weak lensing shows the mass concentration offset from the X-ray emitting gas, coinciding instead with the galaxy (stellar) positions. The standard interpretation is that collisionless dark matter particles passed through the collision while the collisional gas was stripped and left behind.

In the entropy framework, the entropy deficit is tied to the matter distribution, not to the gas specifically. When two clusters collide, the gas (collisional) is stripped by ram pressure and left at the collision center. The galaxies (effectively collisionless) pass through. The entropy-deficit “apparent mass” follows the galaxies, because the entropy balance is evaluated at each point in space relative to the local matter distribution. The lensing signal therefore peaks at the galaxy positions, not at the gas peak—qualitatively reproducing the Bullet Cluster morphology.

The quantitative question is whether the entropy framework produces *enough* apparent mass at cluster scales. Standard MOND underpredicts cluster masses by a factor of ~2 [12]. Whether the causal diamond framework improves on this is an open calculation that requires modeling the entropy balance for cluster-scale mass distributions, which is beyond the scope of this paper. We flag this as the most important open test of the framework (see Section 8).

Part II: Early-Universe Behavior

6. The Requirement

Standard cosmology requires Newtonian (or GR) gravity—without MOND modifications—in at least two early-universe contexts:

The CMB power spectrum. The acoustic oscillations in the CMB, measured to sub-percent precision by Planck [13], are computed using standard GR perturbation theory. Any modification of gravity at the epoch of recombination ($z \approx 1100$) would shift the peak positions and amplitudes, creating a detectable signature. No such signature is observed.

Structure formation. The growth of density perturbations from the CMB epoch to the present requires gravitational collapse governed by standard gravity. MOND modifications at high redshift would alter the growth rate, changing the predicted matter power spectrum.

Any viable MOND framework must therefore reduce to standard gravity at z ■ 1. This is a non-trivial requirement: many MOND formulations treat a_0 as a universal constant that applies at all epochs.

7. The Einstein–de Sitter Limit

In the causal diamond framework, the MOND acceleration at epoch z is [5]:

$$a_0(z) = 2c^2 / [3\pi\, d_{eh}(z)] \qquad (2)$$

where $d_{eh}(z)$ is the proper event horizon at epoch z. In the early universe (z ■ 1), the expansion is matter-dominated and $\Omega_\Lambda(z) \to 0$. In the limit $\Omega_\Lambda = 0$ exactly (the Einstein–de Sitter universe), there is no event horizon: $d_{eh} \to \infty$. Therefore:

$$a_0 \to 0 \quad \text{as} \quad \Omega_\Lambda \to 0 \qquad (3)$$

When $a_0 = 0$, the interpolation function $\mu(x) = kx/(1+kx) \to 1$ for all x, and the MOND equation reduces to $g = g_N$. Gravity is purely Newtonian.

This is the correct behavior. In a universe without dark energy, there is no volume-law entropy to compete with the area-law. All degrees of freedom remain in the Newtonian channel. MOND effects are a consequence of dark energy; where there is no dark energy, there is no MOND.

In our actual universe (Ω_Λ = 0.685), dark energy is present at all epochs but dynamically negligible at high z. The proper event horizon is very large (though finite), and $a_0(z)$ is very small. At z = 1100 (recombination), a_0 is orders of magnitude below any acceleration relevant to CMB physics. The framework is consistent with the CMB.

8. Discussion

8.1 What the framework now addresses

Observational test	Prediction	Status
Galaxy rotation curves	μ(x) = kx/(1+kx), a■ = c²/(2πr_w)	Derived [1, 2]
GW speed (GW170817)	c_gw = c (automatic)	Passed [4]
a■ redshift evolution	47% increase at z = 2	Testable [5]
Gravitational lensing	Lensing = dynamics	Consistent (this work)
CMB / early universe	Standard gravity at z >> 1	Consistent (this work)
Cosmological constant	Λ and a■ from same geometry	Reframed [6]
Bullet Cluster (quantitative)	Enough apparent mass?	OPEN
Cluster mass deficit	Reduce factor-of-2 gap?	OPEN

Table 2. Status of the causal diamond framework across observational tests. Six tests are addressed (four derived, two consistent). Two remain open, both involving galaxy clusters.

8.2 The cluster-scale gap

The two open entries in Table 2 both involve galaxy clusters, where standard MOND underpredicts the observed mass by a factor of ~2 [12]. The causal diamond framework has not yet been applied to cluster-scale mass distributions. Whether the entropy balance at cluster scales produces a larger apparent mass than the point-source MOND formula is a quantitative calculation that requires modeling the entropy deficit for extended, non-spherical mass distributions. This is the most important open test of the framework and the subject of future work.

8.3 Comparison with other MOND frameworks

The causal diamond framework is the only MOND-compatible framework that simultaneously satisfies all of the following: (a) derives a_0 with zero free parameters, (b) derives the interpolation function, (c) predicts $c_{gw} = c$ automatically, (d) predicts lensing = dynamics, and (e) reduces to Newton in the early universe. AeST [14] achieves (c), (d), and (e) but treats a_0 and μ as inputs. TeVeS [9] achieves (d) and (e) but fails (c). No other framework achieves (a) or (b).

9. Conclusion

The causal diamond framework predicts that gravitational lensing tracks dynamical gravity, because the entropy deficit acts as an effective stress-energy source that produces real spacetime curvature. The gravitational action (Einstein-Hilbert) is unmodified, so $c_{gw} = c$; the effective source is modified, so lensing = dynamics. Both follow from the same physical picture: the entropy displacement curves spacetime without changing the rules of wave propagation.

The framework reduces to standard Newtonian gravity in the early universe, because the Einstein–de Sitter limit sends the event horizon to infinity and the volume-law entropy budget to zero. MOND effects are a consequence of dark energy; they vanish when dark energy is negligible. This is consistent with CMB physics and structure formation.

The remaining open challenge is the quantitative performance at cluster scales, where standard MOND falls short by a factor of ~2. Whether the entropy-balance framework closes this gap is the most important calculation that has not yet been done.

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